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The Influence of Multi-Sensory Perception on Public Activity in Urban Street Spaces: An Empirical Study Grounded in Landsenses Ecology

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Abstract: The design of street spaces significantly influences public behavior and quality of life. Understanding how various urban street spatial characteristics affect public behavior, alongside the role of multi-sensory perception, enables designers and planners to create more human-centered urban environments. Grounded in landsenses ecology, this study employs correlation analysis, regression analysis, and Partial Least-Squares Structural Equation Modeling (PLS-SEM) to examine the effects of different urban street spatial characteristics on public behavior and the mediating role of multi-sensory perception. The findings reveal that street spatial characteristics, particularly the Water Surface Ratio (WSR) and Waterfront Density (WD), have a pronounced impact on behavioral traits, with higher public activity frequencies in areas with elevated WSR and WD. Notably, WSR significantly affects static behaviors, such as sunbathing ($\beta = 0.371$, $p < 0.001$), and dynamic behaviors, such as walking ($\beta = 0.279$, $p < 0.001$). While road and water characteristics directly influence behavior, buildings and green spaces mainly affect public behavior through multi-sensory perception. Different sensory perceptions show varying effects, with olfactory perception playing a significant role in these experiences, alongside a notable chain-mediated effect between tactile perception and psychological cognition. These results provide valuable insights for integrating multi-sensory experiences into urban design.

Keywords: landsenses evaluation indices; street space characteristics; PLS-SEM; mediation effect

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Academic Editor: Maria Rosa Trovato

Received: 5 December 2024
Revised: 24 December 2024
Accepted: 27 December 2024
Published: 29 December 2024

Citation: Han, T.; Tang, L.; Liu, J.; Jiang, S.; Yan, J. The Influence of Multi-Sensory Perception on Public Activity in Urban Street Spaces: An Empirical Study Grounded in Landsenses Ecology. *Land* **2025**, *14*, 50. https://doi.org/10.3390/land14010050

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1. Introduction

Cities are confronted with urgent health, social, and environmental challenges, as emphasized by the United Nations Sustainable Development Goals (SDGs) [1]. Urban planning plays a critical role in shaping the spatial structure of cities, influencing lifestyles and environmental exposures that ultimately affect human health and sustainable development [2]. As the concept of "One Health" gains global recognition, the connection between sustainable urban environments and public health becomes increasingly significant. This shift necessitates a focus on human well-being, education, and sustainable living practices [3]. Streets, as vital public spaces within urban life, promote a positive perception that

maintains a continuous relationship between pedestrians and their surroundings [4]. This relationship is essential for enhancing the quality of the living environment.

Urban streets are defined as roadways that facilitate daily social interactions among residents in towns and cities, along with the surrounding buildings, associated vegetation, and other facilities [5]. As fundamental components of the urban built environment, streets significantly influence individual behaviors through their transportation, social, and aesthetic functions [6]. Streets serve as some of the most frequented venues for walking, cycling, and various recreational activities [7]. Research indicates that high-quality street environments effectively promote active engagement among residents [8] and significantly influence public behavior characteristics [9], behavioral satisfaction [10], and route selection [11]. Street greenery significantly impacts both walking frequency and duration [9,12], while elements of the street environment, including pavement and facilities, influence pedestrians' walking preferences [13,14]. Although land use types significantly impact people's behavior, their influence is indirect, typically mediated by the physical characteristics of street space. For instance, wider roads typically incorporate more sidewalks and commercial facilities, which directly facilitate pedestrian activity [15]. In contrast, areas near water offer a peaceful environment, encouraging people to engage in activities such as walking [16]. Therefore, the physical characteristics of street space have a more direct influence on people's behavioral choices. However, most existing studies primarily focus on the relationship between street environments and walking behavior, leading to the insufficient exploration of other behavioral characteristics associated with streets. Previous studies have classified activities into static and dynamic behaviors based on varying levels of physical activity. Static behaviors are characterized by lower levels of physical exertion and energy expenditure, including actions such as resting, reading, conversing, or observing within a specific space [11]. In contrast, dynamic behaviors involve higher levels of physical activity, including spontaneous movements in specific areas, sports, and activities that rely on fitness and recreational facilities [17]. Understanding how the spatial characteristics of streets influence public behavior will help identify effective strategies for enhancing urban environments and improving residents' quality of life.

Various objective environmental characteristics offer the public a range of spatial experiences, triggering complex perceptions and behavior [17]. Scholars have employed the Stimulus–Organism–Response (SOR) model to clarify a series of behavioral responses elicited by the objective environment as a source of stimuli [18]. Landsenses ecology emphasizes individuals' perceptual experiences in urban environments, focusing on the systematic nature of multi-sensory perception and the holistic aspects of psychological cognition. This underscores the necessity of investigating land use planning, construction, and management from multiple related perspectives, including natural elements, physical perception, psychological cognition, socio-economics, processes, and risks [19]. Research indicates that multi-sensory perception is a fundamental condition for enhancing residents' satisfaction in urban environments, while psychological cognition serves as a direct factor influencing this satisfaction [20]. Furthermore, characteristics of visual and auditory perception significantly affect tourists' experiences [21], while sensory perceptions—including visual, auditory, and olfactory—have notable effects on psychological restoration [22,23]. However, existing studies have paid inadequate attention to the impacts of multi-sensory perception and psychological cognition on public activity characteristics. As significant mediators between the environment and behavior, perceptual information influences individuals' cognitive and emotional experiences of space through various channels, including vision, hearing, smell, and touch, thereby directly shaping their behavioral decisions and activity patterns [4]. This influence not only determines the duration of individuals' stays in specific environments, the types of activities they engage in, and their frequency of

activity but also reflects the close relationship between psychological states and the environment. Understanding these mediating roles can aid in designing environments that more effectively meet individuals' sensory needs, thereby enhancing the overall user experience.

Urban street spaces serve as direct facilitators of individuals' perception and understanding of the urban environment. Comprehending the interplay between the components of urban streets and public behavior characteristics is essential for constructing health-centered cities that prioritize human well-being [24]. However, while the characteristics of street spaces, multi-sensory perception, and psychological cognition significantly influence public behavior, comprehensive assessments of their interactions and effects on behavior are still limited. Although some frameworks have identified pathways between physical environmental characteristics and behavioral traits [25,26], few studies have assessed the mediating roles of multi-sensory perception and psychological cognition. Furthermore, existing research often focuses on isolated sensory perceptions. Human sensory perception operates as an integrated system; thus, it is crucial to examine both individual sensory experiences and the cumulative effects of multi-sensory perception. Psychological cognition is fundamentally based on multi-sensory perception, as it synthesizes information from diverse sensory modalities, enabling individuals to form a comprehensive understanding of their environment [27]. This synthesis not only influences emotional responses and decision-making processes but also shapes behavioral outcomes [28,29]. Therefore, examining the mediating roles of multi-sensory perception and psychological cognition is crucial for gaining a deeper understanding of how street spatial characteristics affect public behavior.

This study focuses on street spatial characteristics across four categories: road, building, green space, and water. It quantifies relevant spatial data to explore how these features influence public behavioral traits, with multi-sensory perception and psychological cognition serving as mediators, as illustrated in Figure 1. This research addresses the following scientific questions: (1) What key spatial features influence people's behavioral traits on urban streets? (2) How could multi-sensory perception and psychological cognition shape the pathways through which urban street spatial characteristics impact people's behavior? This study could provide a theoretical foundation for improving urban space design and enhancing residents' quality of life.

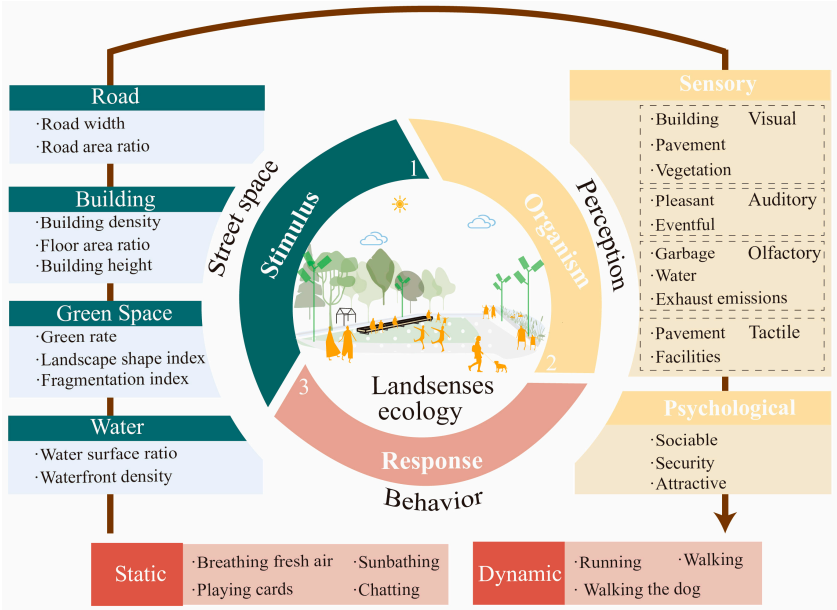


Figure 1. The influence of the spatial characteristics of urban streets on behavioral traits is examined from the perspective of landsenses ecology.

2. Method

2.1. Study Area and Sampling Points

The study was conducted in Fuzhou, a coastal city situated in the eastern part of Fujian Province, China, with a typical subtropical monsoon climate characterized by warm and humid conditions, as well as greenery present throughout the year. The research area is located in Gulou District, a central urban area in the northwest section of Fuzhou integrating various functional areas, such as historical districts, parks, residential areas, and commercial zones. This area features significant landmarks, including Wushan Mountain, the Three Lanes and Seven Alleys historical district, and Xihu Park. Since 2012, the research area has implemented a slow transportation system, with plans for a network that incorporates riverfront, waterfront, lakeside, and park connectivity, aimed at creating a distinctive "Min culture slow life" experience. Considering data availability and the feasibility of field research, this study selects 11 service-oriented streets and 8 landscape-oriented streets as subjects for analysis. The selection criteria for the sites are as follows: (1) The selection of sample sites considered various spatial environmental factors, including roads, buildings, green spaces, and water characteristics, ensuring that the spatial characteristics of the sites varied, thereby guaranteeing their representativeness. (2) To minimize the influence of any single sensory factor (such as traffic noise), the site selection process emphasized a balanced multisensory experience. Finally, based on the field survey, sites that were under construction or deemed unsuitable for research were excluded, resulting in a total of 40 viable research sites, as shown in Figure 2.

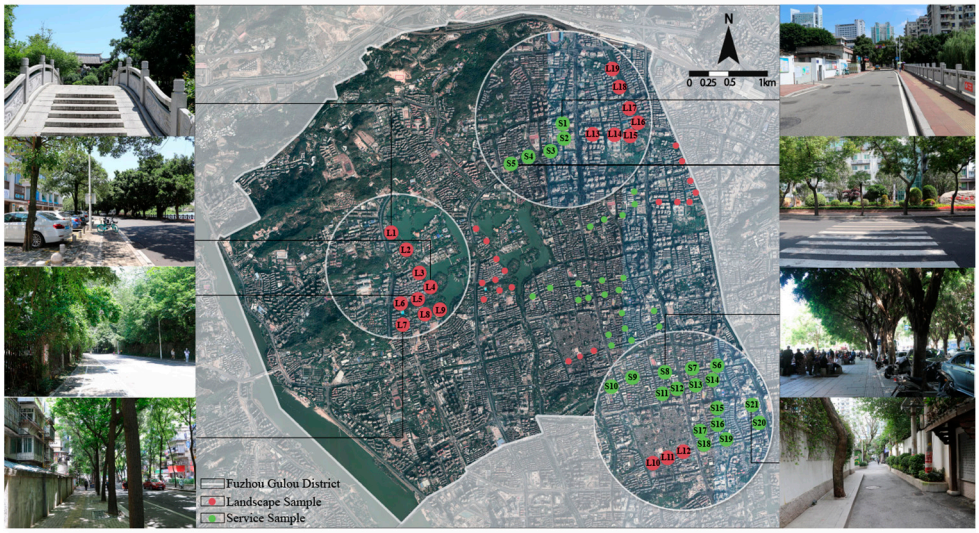


Figure 2. Spatial distribution map of the study area and sample points.

2.2. Data on Street Spatial Features

The research area was categorized into four types of spatial environmental elements—roads (including urban roads, internal roadways, sidewalks, and hard surfaces), buildings, green space, and rivers—using Google satellite imagery of the Gulou District from 2022 (Google Earth). Road network data for the Gulou District of Fuzhou City were obtained from Open Street Map while building outline data and the number of floors were acquired using Local Space Viewer software (version V4.2.0). To analyze the key spatial features influencing landscape perception evaluations of streets in the research area and to account for the scale of sensory perception, buffer zones with a radius of 100 m were established around selected points for survey and indicator calculations [30]. The methods for quantifying urban street space indicators are detailed in Table 1.

Table 1. Quantification of the urban street space indicators.

Elements	Indicators	Calculation Formula	Explanation
Road	Road Width (RW)	$RW = D_v + D_s$	D_v denotes the width of the vehicle lane. D_s represents the width of the sidewalk.
	Road Area Ratio (RAR)	$RAR = \frac{S_r}{S}$	S_r denotes the road surface area within the buffer zone. S denotes the total area within the buffer zone. RAR calculation in this study mainly includes urban main roads and side roads, excluding roads within residential areas.
Building	Building Density (BD)	$BD = \frac{S_g}{S}$	S_g denotes the ground floor area within the buffer zone.
	Floor Area Ratio (FAR)	$FAR = \frac{S_b}{S}$	S_b denotes the total building area within the buffer zone.
	Building Height (BH)	$BH = \frac{\sum_{i=1}^{n} H_i}{n}$	H_i denotes the height of the i-th building within the buffer zone, while n denotes the number of buildings within the buffer zone.
Green Space	Green Rate (GR)	$GR = \frac{S_g}{S}$	S_g denotes the total area of green space within the buffer zone.
	Landscape Shape Index (LSI)	$LSI = \frac{0.25C_g}{\sqrt{S_g}}$	C_g denotes the total perimeter of green space within the buffer zone.
	Fragmentation Index (FI)	$FI = \frac{N_i}{\sum S_i}$	N_i denotes the number of patches within the buffer zone corresponding to the i-th green area, while S_i represents the area of the i-th green space within the buffer zone.
Water	Water Surface Ratio (WSR)	$WSR = \frac{S_w}{S}$	S_w denotes the total area of water within the buffer zone. WSR refers to the proportion of the water surface area within the buffer zone—including lakes, rivers, and artificial ponds—relative to the total area of the buffer zone.
	Waterfront Density (WD)	$WD = \frac{C_w}{S}$	C_w denotes the perimeter of the water's edge within the buffer zone.

2.3. Public Survey Data

2.3.1. Landsenses Evaluation Indices of Urban Streets

Urban streets serve as essential environments for community activities, where environmental factors such as light, sound, smell, color, shape, and texture significantly influence individual perceptions and behaviors. This study establishes an evaluation index system aimed at assessing the perception of urban streets based on their spatial characteristics. Drawing on established landsenses indicators [29,31,32], the system encompasses five dimensions—visual, auditory, olfactory, tactile, and psychological—and integrates landsenses elements associated with roads, buildings, green spaces, and water, as shown in Table 2.

Each of these dimensions is designed to capture specific sensory experiences and their impact on individual psychology. Previous studies typically categorize the street interface into four key components: canopy, building wall, ground plane, and roadside plane [33]. This study primarily focuses on visual perception, particularly factors such as building design, paving material color, and vegetation richness, which collectively shape the street's visual impression and influence pedestrians' emotional responses and behavioral choices. Auditory perception, on the other hand, emphasizes the evaluation of the pleasantness and eventfulness of the street's soundscape [34], particularly focusing on how these auditory

characteristics influence pedestrian behavior patterns. Olfactory perception primarily concerns odors. While positive odors can enhance the overall environmental perception [35], urban streets are frequently exposed to negative smells such as trash, waterborne odors, and vehicle exhaust. This study specifically addresses how these factors significantly influence the perception of street quality. Tactile perception focuses on the physical characteristics of the street, particularly the smoothness of the pavement, the comfort of resting facilities, and their ability to meet pedestrian needs [36]. Street design should focus on optimizing these sensory experiences, providing an environment that is not only safe and comfortable but also attractive, thereby promoting positive pedestrian behavior [33]. In conclusion, the aim of this study is to comprehensively understand how the sensory elements of urban streets shape public behavior and influence the overall perception of space, thereby providing a theoretical basis for optimizing street design.

Table 2. Landsenses indicators of urban street.

Sensory Perception	Characterization Indicators
Visual	The building is colorful. The pavement color is very comfortable. The vegetation is very lush.
Sound	The soundscape is pleasant. The soundscape is eventful.
Olfactory	Odor of the garbage. Odor of the water. Odor of automobile exhaust.
Tactile	The pavement is smooth. The rest facilities are well-equipped. The rest facilities are comfortable.
Psychological	The atmosphere of interaction is well. It has a sense of security. It is attractive.

2.3.2. Questionnaire Design

The public survey data for this study were collected using a structured questionnaire. The questionnaire consisted of three main sections. The first section gathered basic demographic information about the respondents. The second section focused on the evaluation of sensory experience and psychological perception related to the spatial characteristics of the streets. Drawing on the previously established evaluation index system for landsenses ecology, respondents were instructed to assess their environment after a one-minute observation across five dimensions: visual, auditory, olfactory, tactile, and psychological, using a scale ranging from 1 (strongly disagree) to 7 (strongly agree). The third section focused on the types and frequencies of the respondents' activities. Based on observations of the behaviors of street users, these activities were classified as static or dynamic, as shown in Table 3. Respondents were asked to indicate the frequency of their activities using a seven-point Likert scale, ranging from 1 (rarely) to 7 (very frequently). The survey utilized a random sampling method. Prior to participation, respondents were thoroughly informed of the study's objectives and gave their consent. Participants were subsequently guided through the completion of the questionnaire. To ensure the representativeness of the sample, respondents were selected from a broad spectrum of demographic backgrounds, including gender, age, and occupation, thereby improving the generalizability and applicability of the results. The random selection and demographic diversity of the sample bolstered the reliability and validity of the study's findings. To account for crowd

activity and ensure the feasibility of the survey, data collection was conducted on clear weather weekdays during three specific time periods: from 8:00 to 11:00 a.m., from 2:00 to 5:00 p.m., and from 5:00 to 8:00 p.m. Ultimately, a total of 471 valid questionnaires were collected from 40 sampling sites.

Table 3. Classification of behavior types in urban streets.

Behavior Type	Concrete Behavior
Static activities	breathe the fresh air, sunbathe, chat, play cards.
Dynamic activities	run, walk, walk the dog, shopping

2.4. Construction of the Conceptual Model

This study investigates four categories of street spatial characteristics: buildings, roads, green spaces, and water. From the perspective of landsenses ecology, this research aims to quantify individuals' multi-sensory perception and psychological cognition while investigating the mechanisms through which street spatial characteristics influence behavior. The model is designed to illustrate the pathways through which the four categories of street spatial characteristics impact the two dimensions of activity types, with street spatial characteristics serving as independent variables, behavioral traits as dependent variables, and multi-sensory perception and psychological cognition as mediators. A structural equation model is proposed, as illustrated in Figure 3, along with the following specific research hypotheses:

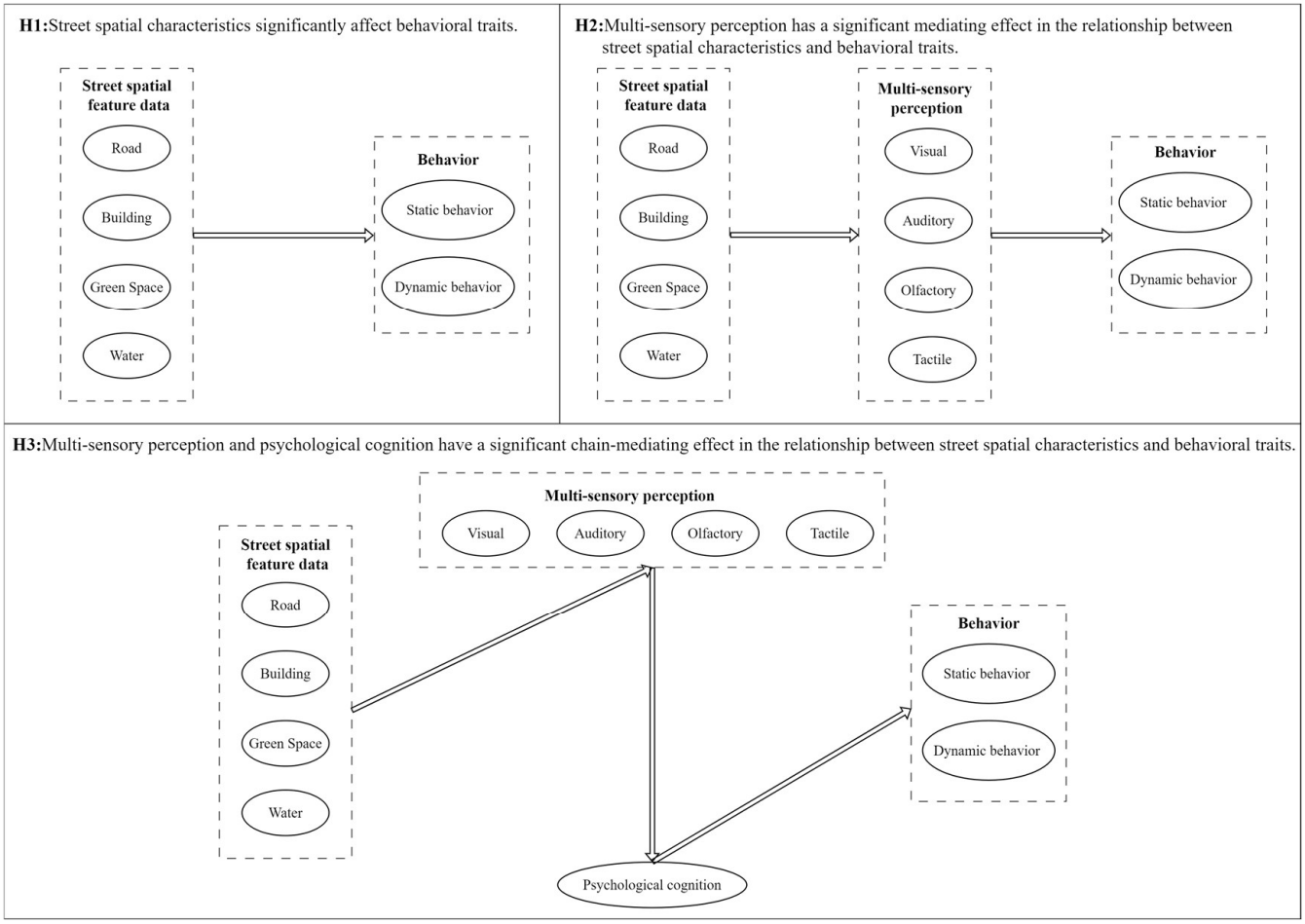


Figure 3. Impact pathways of urban street spatial characteristics on behavioral outcomes.

H1: Street spatial characteristics significantly affect behavioral traits.

H2: Multi-sensory perception has a significant mediating effect in the relationship between street spatial characteristics and behavioral traits.

H3: Multi-sensory perception and psychological cognition have a significant chain-mediating effect in the relationship between street spatial characteristics and behavioral traits [26,29].

This study employed Partial Least Squares-Structural Equation Modeling (PLS-SEM) as the statistical technique. PLS-SEM is utilized to test the hypotheses, offering distinct advantages over traditional structural equation models, particularly for smaller sample sizes and exploratory research [37]. This multivariate analysis comprises two interconnected components: the measurement model, which assesses reliability and validity to confirm the predictive potential of observed variables for their latent constructs, and the structural model, which evaluates the predictive relationships among these constructs through hypothesis testing and mediation analysis. The sample size for this study (N = 471) exceeds the recommended minimum of ten times the number of questionnaire items (32 items), thereby ensuring reliable outcomes from the PLS-SEM analysis. Although PLS-SEM provides substantial advantages, several limitations need to be addressed. Firstly, even though the sample size surpassed the recommended threshold, its representativeness might still restrict the broader applicability of the findings [37]. Moreover, PLS-SEM necessitates a balance between sample size adequacy and the robustness of standardized model evaluation criteria, which highlights areas requiring methodological refinement [38]. Statistical analysis was carried out using Smart PLS 3.0.

3. Results

3.1. Basic Characteristics of Urban Street Spatial Features

The spatial characteristics of the streets in the study area are detailed in Table 4. The average Road Width (RW) is 13.9 m, with the majority of streets designed with two lanes to accommodate bidirectional traffic. The average Building Height (BH) is approximately 24 m, primarily comprising seven-story residential structures, and the average Building Density (BD) is 28.5%, reflecting an appropriate level of land use intensity. The average Green Rate (GR) is 35.5%, and the low fragmentation value suggests that the green spaces in the study area are relatively intact, indicating a high level of green space development. Few roads are located near rivers, and both the Water Surface Ratio (WSR) and Waterfront Density (WD) are low.

Table 4. Statistical analysis of street space features.

Elements	Indicators	Mean	Max	Min	Standard Deviation
Road	RW/m	13.9	20.0	8.3	3.0
	RAR/%	41.9	66.6	12.5	13.4
Building	BD/%	28.5	45.4	8.6	8.7
	FAR	2.017	4.843	0.196	1.075
	BH/m	24.499	50.267	5.279	9.542
Green Space	GR/%	35.5	64.2	20.5	12.0
	LSI	11.367	20.732	5.417	3.713
	FI/%	0.106	0.299	0.009	0.074
Water	WSR/%	0.079	0.345	0.000	0.113
	WD/%	0.856	3.642	0.000	1.121

3.2. Sample Statistics

As shown in Figure 4, among the 471 valid responses, 267 (57%) are male, while 204 (43%) are female, yielding a gender ratio of approximately 1:1.3. The largest proportion of respondents falls within the age group of 25 to 30 years, comprising 21% of the total. Furthermore, 77% of respondents indicate that their commuting time to the survey location does not exceed 0.5 h.

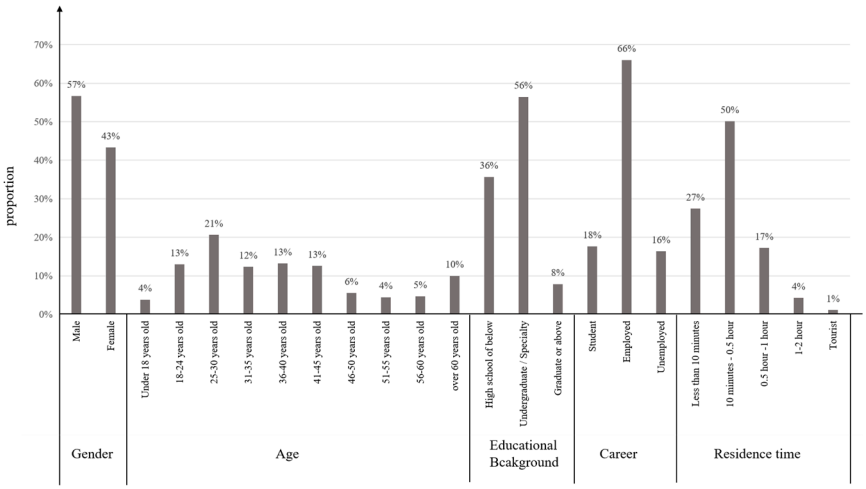


Figure 4. Composition analysis of the respondent group.

Individual characteristics (such as gender, age, etc.) are potential factors contributing to differences in the evaluation of street perception indicators. Given the non-normal distribution of the data, non-parametric tests are employed to investigate the difference in perception indicators among groups with varying individual characteristics and to identify statistically significant variations. The results, as summarized in Table 5, show that gender exhibits significant differences in Well-equipped facilities, Interaction, and Security, with males assigning higher scores than females. No significant differences are observed across age groups for any perception indicators. Educational background demonstrates significant differences in the evaluation of Sound Pleasantness, Garbage Odors, and Automobile Exhaust Odors, indicating variations in how individuals with different educational backgrounds evaluated these aspects. Similarly, people with different occupations show different opinions on Garbage Odors, Well-equipped Facilities, and Interaction, with unemployed individuals assigning lower scores to Garbage Odors but higher scores to Well-equipped Facilities and Interaction compared to students and employed individuals. Finally, residential distance shows significant differences in the perception of "the pavement color is very comfortable", with non-local visitors and those living within 30 min of the sampled street providing the highest ratings.

Individual characteristics also show significant variation in their association with public behavioral patterns in street environments. As presented in Table 6, gender shows significant differences in static behaviors, including chatting, playing cards, and sunbathing, with males engaging in these activities more frequently than females. Age shows notable differences across all behaviors except chatting and shopping, with individuals aged 51–60 being the most active group across most activities. Educational background exhibits significant differences across all behaviors. Respondents with undergraduate or specialty degrees are the most active ones in shopping, while those with higher education levels participate more frequently in other activities. Conversely, residential distance from the sampled site does not reveal any significant difference in behavioral patterns.

Table 5. Summary of the non-parametric test results examining the relationships between individual characteristics and landsenses indicators.

Indicator		Gender	Age	Educational Background	Career	Residence Time
Visual	Building	0.054	0.086	0.312	0.262	0.378
	Pavement	0.711	0.212	0.847	0.243	0.004 **
	Vegetation	0.194	0.576	0.247	0.192	0.851
Sound	Pleasantness	0.358	0.268	0.027 *	0.052	0.488
	Eventfulness	0.558	0.990	0.146	0.842	0.476
Olfactory	Garbage	0.590	0.094	0.024 *	0.012 *	0.878
	Water	0.632	0.316	0.381	0.936	0.521
	Automobile	0.64	0.263	0.042 *	0.871	0.360
Tactile	Pavement	0.551	0.101	0.058	0.171	0.177
	Well-equipped facilities	0.039 *	0.091	0.063	0.033 *	0.878
	Comfortable facilities	0.268	0.274	0.186	0.440	0.787
Psychological	Interaction	0.045 *	0.125	0.318	0.050 *	0.748
	Security	0.007 **	0.349	0.207	0.488	0.712
	Attraction	0.257	0.247	0.187	0.076	0.279

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table 6. Summary of non-parametric test results analyzing the relationships between individual characteristics and behavioral frequencies.

Behavior		Gender	Age	Educational Background	Career	Residence Time
Static activity	breath the fresh air	0.998	0.004 **	0.001 **	0.000 ***	0.287
	chat	0.024 *	0.138	0.001 **	0.575	0.840
	play cards	0.026 *	0.002 **	0.000 ***	0.000 ***	0.487
	sunbathe	0.008 **	0.011 *	0.000 ***	0.005 **	0.460
Dynamic activities	walk the dog	0.952	0.006 **	0.005 **	0.015 *	0.501
	run	0.282	0.015 *	0.012 *	0.092	0.348
	walk	0.829	0.002 **	0.012 *	0.000 ***	0.552
	shopping	0.441	0.061	0.032 *	0.024 *	0.668

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

3.3. Relationship Between Street Spatial Characteristics and Behavioral Traits

As shown in Figure 5, regarding road characteristics, Road Width (RW) is positively correlated with activities such as enjoying fresh air, sunbathing, and walking but negatively correlated with shopping. The Road Area Ratio (RAR) is negatively correlated exclusively with running, while it is positively correlated with shopping. In terms of building characteristics, Building Density (BD) does not show a significant correlation with chatting; however, it is significantly negatively correlated with other behaviors, except for shopping. The correlation results for the Floor Area Ratio (FAR) align with those of Building Density (BD), indicating no significant correlation with either chatting or sunbathing. The average Building Height (BH) is significantly positively correlated only with chatting and sunbathing. Concerning green space, the Green Rate (GR) is positively correlated with running but shows a significant negative correlation with both chatting and shopping. The Landscape Shape Index (LSI) demonstrates a positive correlation with chatting, walking, and shopping behaviors. The Fragmentation Index (FI) is negatively correlated with activities including enjoying fresh air, sunbathing, walking dogs, running, and walking, while it shows a positive correlation with shopping. Regarding water, both the Water Surface

Ratio (WSR) and Waterfront Density (WD) are significantly positively correlated with all behaviors except shopping, which shows a negative correlation.

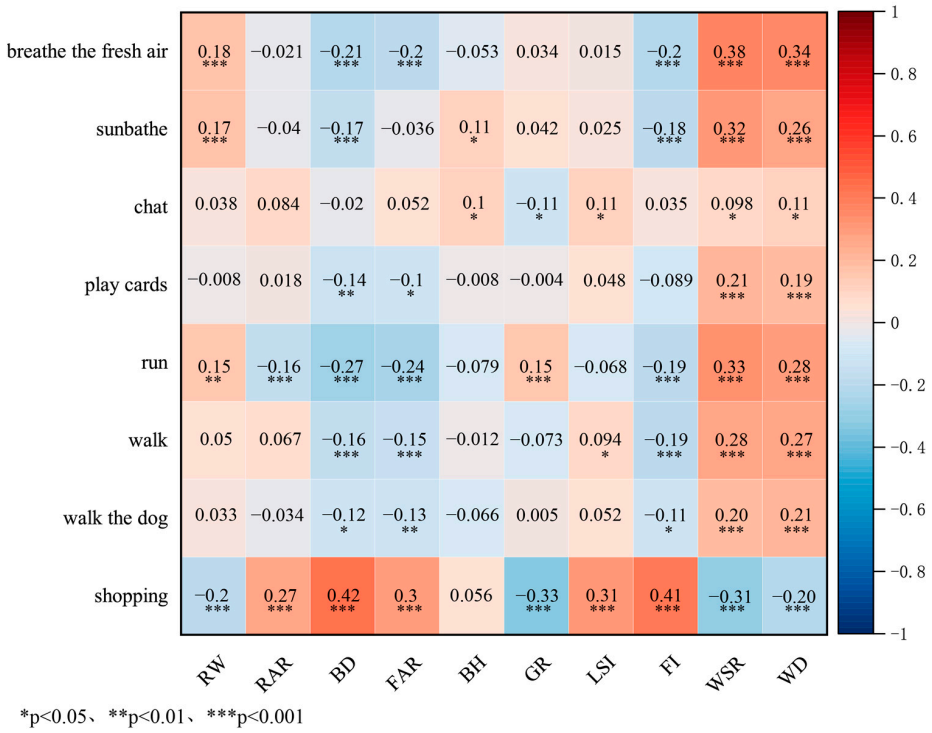


Figure 5. Correlation analysis results between the spatial characteristics of urban streets and behavioral traits.

3.4. Influence of Street Spatial Characteristics on Behavior Traits

3.4.1. Factors Influencing the Impact of Street Spatial Characteristics on Behavioral Traits

To further investigate the influence of street spatial characteristics on behavior, a multiple regression analysis is conducted, treating behavioral traits as the dependent variable and street spatial characteristics as the independent variable, as shown in Table 7. The results indicate that water characteristics could significantly influence the frequency of public behavior. Road characteristics have a limited effect, influencing only the activities associated with enjoying fresh air and shopping. Building characteristics are related to activities such as enjoying fresh air, running, walking, and shopping. Green space could significantly affect the frequency of all behaviors, except for dog walking and running.

Table 7. Multiple stepwise regression analysis of street spatial characteristics and behavior.

Type	Dependent Variable	Independent Variable	*Beta*	*VIF*	*t*	R^2	*F*
Static activity	Breathe the fresh air	RW	0.138	1.479	2.716 **	0.186	21.226 ***
		FAR	−0.137	1.68	−2.525 *		
		GR	−0.186	1.41	−3.738 ***		
		WSR	0.213	3.134	2.884 **		
		WD	0.147	2.508	2.214 *		
	chat	GR	−0.134	1.005	−2.944 ***	0.039	9.397 ***
		WD	0.154	1.005	3.387 **		
	play cards	WD	0.268	1.000	6.020 ***	0.072	36.242 ***
	sunbathe	GR	−0.116	1.073	−2.588 **	0.129	34.533 ***
		WSR	0.371	1.073	8.299 ***		

Table 7. *Cont.*

Type	Dependent Variable	Independent Variable	*Beta*	*VIF*	*t*	R^2	*F*
Dynamic activities	walk the dog	WD	0.229	1.000	5.102 **	0.053	26.028 ***
	Run	BD	−0.206	1.159	−4.388 ***	0.108	2.128 ***
		WD	0.191	1.159	4.059 ***		
	walk	FAR	−0.194	1.415	−3.823 ***	0.150	27.500 ***
		GR	−0.284	1.261	−5.92 ***		
		WSR	0.279	1.215	5.939 ***		
	shopping	RAR	0.149	1.261	3.239 **	0.218	32.386 ***
		BD	0.198	2.287	3.2 **		
		GR	−0.129	1.497	−2.58 *		
		WSR	−0.14	1.793	−2.556 *		

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

3.4.2. The Impact Pathways of Street Spatial Characteristics on Behavioral Traits

When evaluating the Structural Equation Model (SEM), the quality of the model is primarily assessed through reliability and validity tests. The evaluation typically involves Composite Reliability (CR) and Average Variance Extracted (AVE) for each latent variable to gauge the model's reliability and validity [37]. After normalizing the data, we conducted reliability and validity analyses and found that the CR coefficients exceeded 0.8, and the AVE values were all above 0.6. These results indicate that the model demonstrates good internal consistency and convergent validity.

To meet the requirements for structural validity, the factor loadings of all indicators (representing the influence of observable variables on latent variables) must exceed 0.55 [37,39]. However, the factor loadings for the RAR, BH, LSI, and FI are below this threshold. To enhance the model's validity, we consider excluding these indicators. Additionally, the indicators for walking and shopping do not meet the established criteria and are subsequently removed from the model.

In terms of direct effects, only road and water characteristics significantly influence both static and dynamic behaviors. As shown in Table 8, the 95% confidence intervals for the estimated indirect effects do not include zero, indicating that there is a mediating effect of multi-sensory perception within this relationship.

Table 8. Results of the mediation effects of multi-sensory perception on the relationship between street spatial characteristics and behavioral traits.

Dependent Variable	Independent Variable	Direct Effect			Indirect Effect		
		Estimates (β)	*p*	95%CI	Estimates (β)	*p*	95%CI
Static activity	Road	−0.091	0.023	[−0.172,−0.015]	0.141	0.001	[0.098,0.192]
	Building	0.032	0.652	[−0.108,0.169]	−0.087	0.008	[−0.155,−0.024]
	Green space	−0.075	0.080	[−0.158,0.010]	−0.095	0.001	[−0.151,−0.044]
	Water	0.192	0.001	[0.081,0.304]	0.176	0.001	[0.109,0.248]
Dynamic activities	Road	−0.100	0.022	[−0.187,−0.013]	0.097	0.000	[0.057,0.144]
	Building	−0.053	0.470	[−0.199,0.090]	−0.054	0.040	[−0.103,−0.001]
	Green space	0.020	0.716	[−0.083,0.129]	−0.055	0.015	[−0.101,−0.011]
	Water	0.127	0.039	[0.004,0.246]	0.121	0.001	[0.058,0.198]

Figure 6 illustrates the significant pathways of indirect effects. Roads exert a direct influence on both static and dynamic behaviors, alongside a positive indirect effect mediated by auditory, olfactory, and tactile perceptions. Although buildings do not exhibit a

significant direct effect on static and dynamic behaviors, they negatively influence these behaviors indirectly through tactile perception. Green spaces do not significantly impact static and dynamic behaviors; however, they negatively affect static behaviors through auditory and tactile perceptions and have a negative influence on dynamic behaviors via tactile perception. Water has a direct effect on both static and dynamic behaviors, providing a positive indirect influence on static behaviors through auditory, olfactory, and tactile perceptions, as well as a positive indirect effect on dynamic behaviors through olfactory and tactile perceptions.

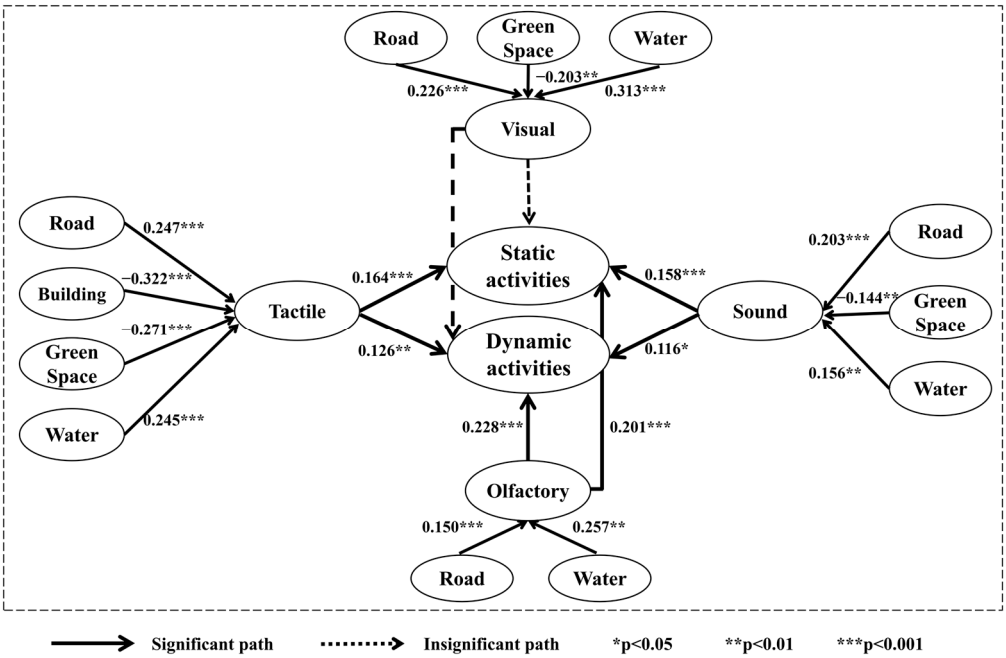


Figure 6. Influence pathways of urban street spatial characteristics on behavioral traits.

As shown in Table 9, further investigation into the mediating roles of multi-sensory perception and psychological cognition demonstrates that street spatial characteristics exert a chain-mediated effect on static behaviors through tactile perception and psychological cognition. Additionally, water influence psychological cognition through visual perception, which subsequently affects static behaviors.

Table 9. Significant pathways of chain mediation effects.

Influence Path	Estimates (β)	*p*	95%CI
Road -> tactile perception -> psychological cognition -> static activity	0.014	0.015	[0.004,0.027]
Building -> tactile perception -> psychosocial cognition -> static activity	−0.019	0.015	[−0.035,0.005]
Green space -> tactile perception -> psychological cognition -> static activity	−0.016	0.018	[−0.031,−0.005]
Water -> tactile perception -> psychological cognition -> static activity	0.014	0.025	[0.004,0.028]
Water -> visual perception -> psychological cognition -> static activity	0.006	0.045	[0.001,0.013]

4. Discussion

4.1. The Relationship Between Street Spatial Characteristics and Behavioral Traits

Street spatial characteristics significantly influence public behavioral traits, with the presence of water exerting a particularly pronounced effect on activity levels. Environments featuring water are generally more appealing to the public [40], aligning with prior research

that indicates spaces near water are more attractive and that water plays a significant role in influencing static behaviors in parks [16,17]. Conversely, the proportion of green space has a notable negative impact on public behavior. Previous studies have shown that the influence of urban green spaces is nonlinear; for instance, the visibility of green streetscapes and NDVI levels exhibit nonlinear effects on walking tendencies [41,42]. Moreover, the quantity of green spaces has a significant inverted U-shaped effect on life satisfaction, with an optimal proportion of 11% maximizing public life satisfaction [43]. In our case, the Green Rate (GR) ranges from 20.5% to 64.2%, which falls within the declining segment of this threshold. Therefore, while increasing green space may continue to positively influence behavior, exceeding the optimal ratio could diminish these beneficial effects and potentially result in negative outcomes. The adverse impact of the Green Rate (GR) is more pronounced on static behaviors, consistent with previous findings that visitors prefer lower per capita green space when seated compared to when walking [44].

According to the regression analysis results, both Road Width (RW) and Road Area Ratio (RAR) significantly and positively influence activity frequency. This supports earlier research indicating that Road Width (RW) is a critical variable influencing street activity [45], significantly impacting the fulfillment of walking needs [46], and showing a strong positive correlation with social interactions among middle-aged and elderly individuals [47]. Apart from shopping, building characteristics correlate negatively with behavior. Previous studies suggest a nonlinear relationship between Building Density (BD) and pedestrian traffic; specifically, in urban centers where Building Density (BD) exceeds 0.3, building characteristics negatively impact walking [48]. Future research should concentrate on the specific effects of spatial characteristics on behavior in high-density environments to enhance public activity experiences.

4.2. Pathways Through Which Street Spatial Characteristics Influence Behavior

Multi-sensory perception plays a crucial mediating role in the relationship between street spatial characteristics and behavioral traits. Specifically, roads and water not only directly influence both static and dynamic behaviors but also exert an indirect effect through multi-sensory perception. Conversely, building and green space characteristics primarily impact static and dynamic behaviors via multi-sensory perception. Previous research indicates that soundscapes significantly affect pedestrians' walking routes and behavioral patterns [49], influencing walking speed and extending dwell time [50,51]. Similarly, odors can affect crowd movement speed and dwell time [51]. Furthermore, pavement evenness, as well as the number and placement of resting facilities, are critical factors influencing outdoor activities [52]. These findings collectively underscore the significant influence of multi-sensory perception on behavior.

However, existing studies have primarily focused on the role of multi-dimensional physical perception in psychological restoration, with limited exploration of its effects on behavior. Prior research has highlighted the significant impact of visual, auditory, and olfactory senses on public restoration [22,23,53]. Among these, olfactory perception is particularly vital in the restoration process [53], supporting this study's conclusion that olfactory perception has the most substantial impact on behavioral activities. Humans possess a remarkable sense of smell, often surpassing that of many animals [54]. Research has demonstrated that in urban forest parks, children's olfactory experiences have the most significant influence on their behavior [55]. Moreover, when examining the effects of multi-sensory stimulation in urban green spaces on physiological stress, olfactory stimuli have been found to be more effective than visual stimuli in alleviating stress [56].

In the pathway of "street spatial characteristics—sensory perception—psychological cognition—behavioral traits", roads, buildings, green spaces, and water influence psycho-

logical cognition through tactile perception, which subsequently affects static behaviors. Tactile perception and psychological cognition serve as chain mediators in this process. Evidence indicates that multi-sensory stimulation from pleasant landscape features can lead to positive cognitive evaluations [28,29], thereby enhancing public behavior. Research from the perspective of landsenses ecology emphasizes the significance of tactile perception indicators [32,57]. For hearing-impaired visitors, tactile experiences are notably richer than those from other non-visual senses [58]. Furthermore, studies on the restorative effects of multi-sensory perception in urban green spaces suggest that tactile perception has an indirect relationship with psychological restoration, influencing residents' behavior and emotional responses, which in turn affects restoration outcomes [59]. This implies a closer connection between tactile perception and emotional cognition compared to other senses. In this study, individuals develop positive tactile perceptions through interactions with various paving materials and the use of resting facilities, fostering a strong cognitive atmosphere that influences behaviors such as playing cards and chatting. Thus, tactile perception plays a crucial role in understanding the environment and interacting with surrounding elements.

4.3. Research Limitations and Future Studies

While this study primarily focuses on the impact of street spatial characteristics on public behavior, land use and social activities may indirectly influence these behaviors by shaping spatial functions and patterns of human mobility. Future research could examine how these factors intersect with sensory experiences to shape public activity patterns in urban streets. Moreover, this study does not fully address the potential impacts of seasonal climate variations on pedestrian behavior and street space usage. Variations in temperature and humidity, in particular, may affect pedestrian comfort and, consequently, alter the types and frequency of activities in urban streets. Future studies could investigate how these climatic variations interact with human behavior and spatial usage, offering deeper insights into how climate shapes public engagement with urban environments.

5. Conclusions

Urban streets serve as essential public spaces that connect individuals to their environment, with spatial characteristics playing a pivotal role in shaping public perception and behavior. This study explores the influence of street spatial characteristics on public behavioral traits, emphasizing the mediating role of multi-sensory perception. Grounded in landsenses ecology, the research employs methodologies such as spatial analysis, correlation analysis, regression analysis, and Structural Equation Modeling (SEM). The key findings are summarized as follows:

(1) Among street spatial characteristics, water elements have the most substantial impact on public behavioral traits. Areas with higher Water Surface Ratios (WSRs) promote nature-based activities, while higher Waterfront Density (WD) fosters social and physical activities, such as chatting, running, and walking dogs.

(2) Roads and water features influence behaviors directly and indirectly through multi-sensory perception, particularly auditory, olfactory, and tactile modalities. Conversely, buildings and green spaces affect behaviors primarily through sensory pathways, with certain characteristics, such as Building Density (BD) and Green Rate (GR), having negative impacts.

(3) Olfactory perception surpasses other sensory modalities in influencing public behavior. Unpleasant odors significantly deter public engagement, even in visually and acoustically appealing areas, highlighting the importance of managing odor sources in urban environments.

This study underscores the need to integrate multi-sensory perception into urban planning to enhance public engagement and well-being. Special attention should be given to olfactory landscapes and the interplay between tactile perception and psychological cognition. By optimizing these sensory dimensions, urban spaces can become safer, more attractive, and better connected to the communities they serve.

Author Contributions: Conceptualization, T.H., L.T. and J.L.; methodology, T.H. and J.L.; software, T.H.; validation, T.H., S.J. and J.Y.; formal analysis, T.H.; investigation, T.H. and S.Y; resources, J.L.; data curation, T.H.; writing—original draft preparation, T.H.; writing—review and editing, L.T., J.L., S.J. and J.Y.; visualization, T.H.; supervision, L.T. and J.L; project administration, J.L.; funding acquisition, J.L. All authors have read and agreed to the published version of the manuscript.

Funding: This work was supported by the National Natural Science Foundation of China (NO.52378049).

Data Availability Statement: Due to the privacy and confidentiality of the respondents, the questionnaire data used in this study cannot be made publicly available. However, the data can be accessed upon reasonable request and under the condition that participant privacy is ensured, by contacting the corresponding author.

Acknowledgments: We express gratitude to two individuals for their contributions: Master's student Mengqiao Zhang and Shiying Li from Fuzhou University provided assistance in data collection and article revision.

Conflicts of Interest: The authors declare no conflicts of interest.

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